

Mid Term Exam Questions Spring 25

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Please submit your questions by Friday, 14 March 2025. Please make sure you make Set A and Set B of your question. Please provide a handwritten/latex solution with the marks distribution to your question.

- While posting your question here, please follow this format:

(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

CO1 Mandatory Question (ZAZ):

Set A

CO1	Implement the logic function, $Y = \overline{A} + \overline{B}C$ using any combination of the logic families you know. Then, mention the operating mode of your diodes/transistors when the output is low . <i>[You may simplify the function if possible. Mentioning the values of circuit components is not necessary.]</i>	[10]
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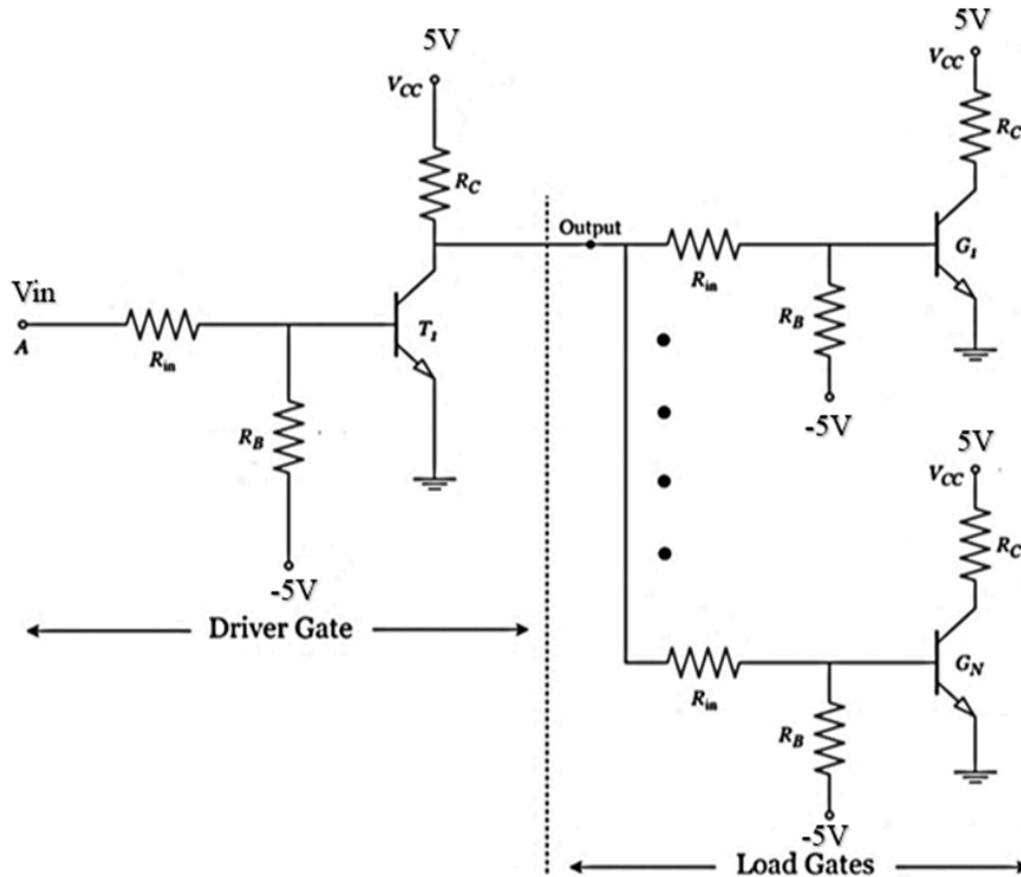
Set B

CO1	Implement the logic function, $Y = \overline{A + B} \cdot \overline{C}$ using any combination of the logic families you know. Then, mention the operating mode of your diodes/transistors when the output is high . <i>[You may simplify the function if possible. Mentioning the values of circuit components is not necessary.]</i>	[10]
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RTL (NUR):

Set-A

For the given circuit assume $V_{OH} = 4.5V$ and $V_{OL} = 0.2V$. Also assume common emitter current gain, $\beta_F = 30$. $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.2V$ and cut in voltage for transistor $V_\gamma = 0.5V$.



Driver and Loads are identical where $R_{in} = 5k$, $R_B = 100k$, and $R_C = 2k$.

(a)	CO2	If $V_{in} = \text{High}$, find the power dissipation in the Driver circuit assuming 48 loads are connected.	10
(b)	CO2	Find the value of Noise margin of the circuit.	10

Set-A (Soln)

(a) $V_{in} = \text{high} = 5V \rightarrow T_1 \rightarrow \text{saturation}$

$$V_B = 0.8V, V_C = 0.2V$$

$$I_1 = \frac{5 - 0.8}{5k} = 0.84mA$$

$$I_2 = \frac{0.8 - (-5)}{100k} = 0.058mA$$

$$I_B = I_1 - I_2 = 0.782mA$$

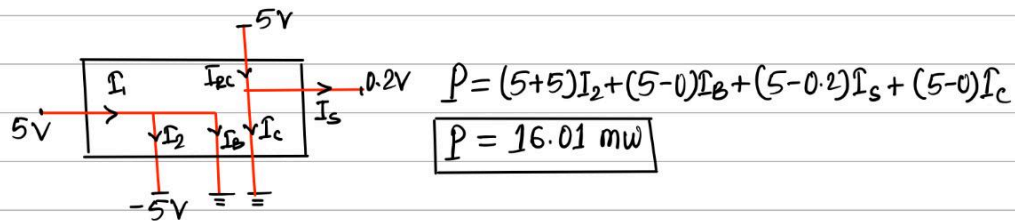
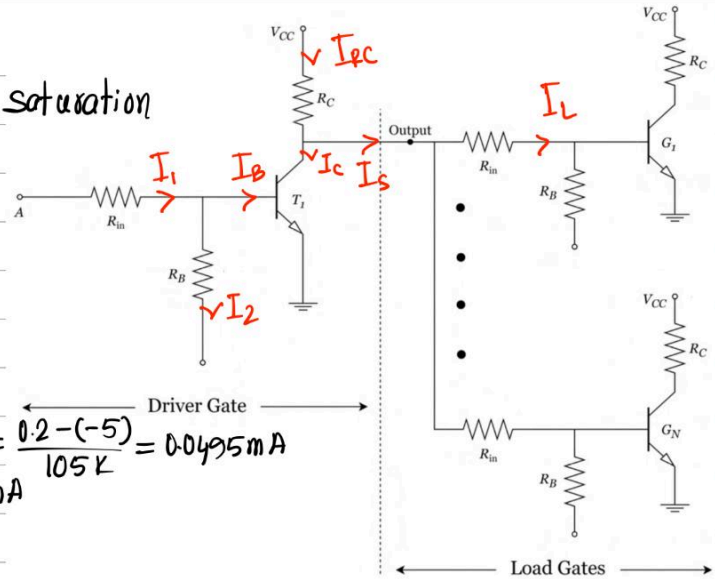
Load Transistors $\rightarrow$ cutoff

$$\text{individual load current, } I_L = \frac{0.2 - (-5)}{105k} = 0.0495mA$$

$$\text{total } I_S = 48 I_L = 2.376mA$$

$$I_{EC} = \frac{5 - 0.2}{2k} = 2.4mA$$

$$\therefore I_C = I_{EC} - I_S = 0.024mA$$



(b) V_{IL} calculation

T_1 is on the verge of turn on $\rightarrow V_B = V_T = 0.5V$

$$I_2 = \frac{0.5 + 5}{100k} = 0.055mA$$

$$I_1 = I_2 = 0.055 = \frac{V_i - 0.5}{5k}$$

$$\Rightarrow V_i = 0.775V$$

$$\therefore V_{iL} = 0.775V, V_{OL} = 0.2V \rightarrow V_{NL} = V_{iL} - V_{OL} = 0.575V$$

V_{IH} Calculation

$T_1 \rightarrow \text{sat}$, on the verge of saturation and f. Active.

$$\therefore \beta_{forced} \approx \beta_F = 30, V_B = 0.8V, V_C = 0.2V$$

$$I_C = \frac{5 - 0.2}{2k} = 2.4mA, \beta = \frac{I_C}{I_B} \Rightarrow I_B = \frac{I_C}{30} = 0.08mA$$

$$I_2 = \frac{0.8 + 5}{100k} = 0.058mA, I_1 = I_2 + I_B = 0.138mA$$

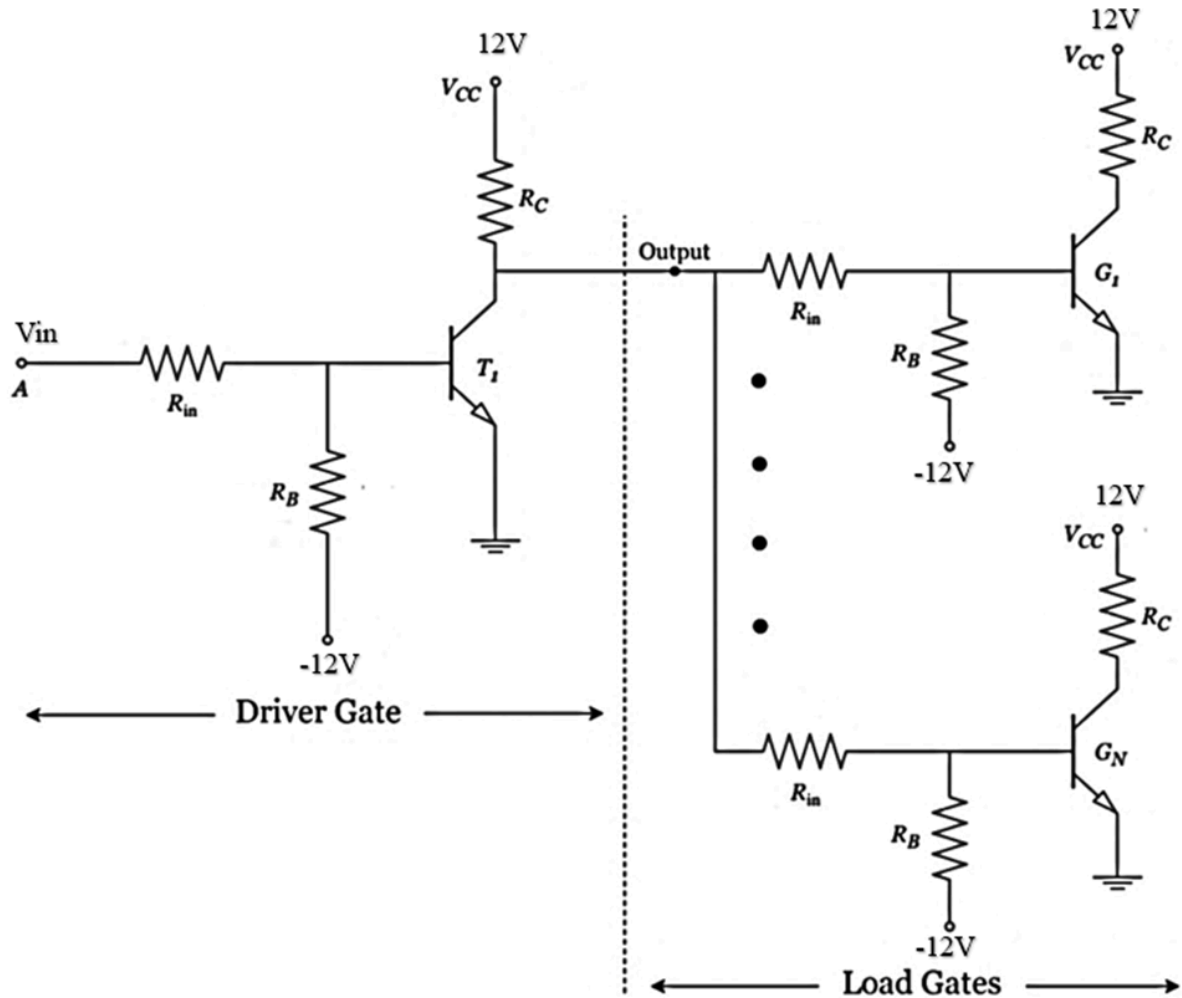
$$I_1 = 0.138 = \frac{V_i - 0.8}{5k} \Rightarrow V_i = 1.5V$$

$$\therefore V_{IH} = 1.5V, V_{OH} = 4.5V \therefore V_{NH} = 4.5 - 1.5 = 3V$$

$$\therefore \text{Noise Margin} \rightarrow NM = \min(V_{NL}, V_{NH}) = 0.575V$$

Set-B

For the given circuit assume $V_{OH} = 10V$ and $V_{OL} = 0.2V$. Also assume common emitter current gain, $\beta_F = 30$. $V_{BE}(\text{sat}) = 0.8V$, $V_{CE}(\text{sat}) = 0.2V$ and cut in voltage for transistor $V_\gamma = 0.5V$.



Driver and Loads are identical where $R_{in} = 15k$, $R_B = 100k$, and $R_C = 2.2k$.

(a)	CO2	If Vin = High , find the power dissipation in the Driver circuit assuming 50 loads are connected.	10
(b)	CO2	Find the value of Noise margin of the circuit.	10

Set-B (Soln)

(a) $V_{in} = \text{high} = 12\text{V} \rightarrow T_1 \rightarrow \text{saturation}$

$$V_B = 0.8\text{V}, V_C = 0.2\text{V}$$

$$I_1 = \frac{12 - 0.8}{15\text{K}} = 0.747\text{mA}$$

$$I_2 = \frac{0.8 - (-12)}{100\text{K}} = 0.128\text{mA}$$

$$I_B = I_1 - I_2 = 0.619\text{mA}$$

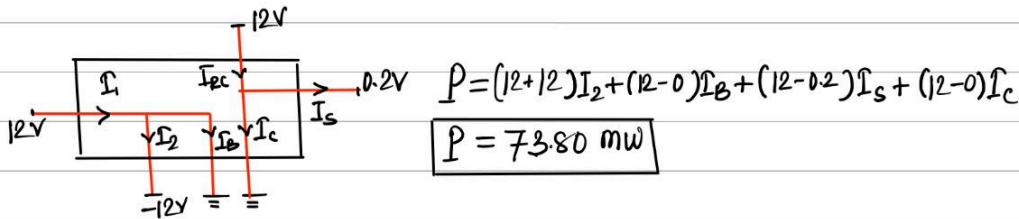
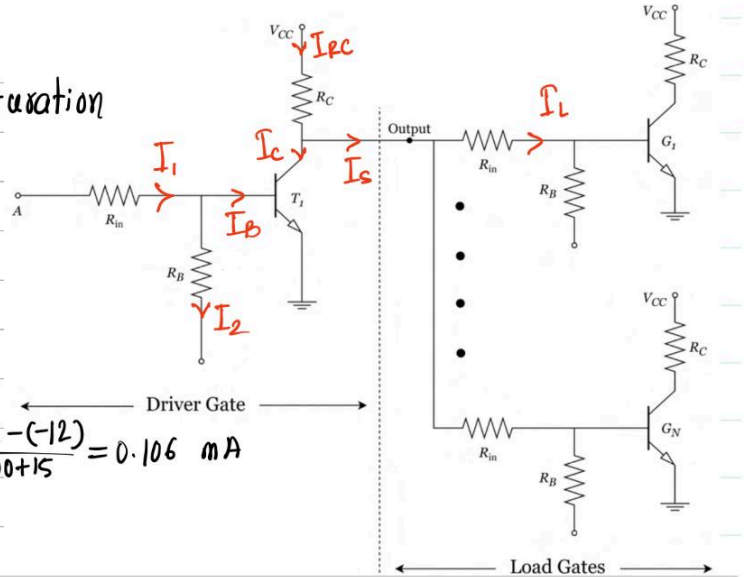
Load Transistors $\rightarrow$ Cutoff

$$\text{individual load current, } I_L = \frac{0.2 - (-12)}{100 + 15} = 0.106\text{mA}$$

$$\text{total } I_S = 50 I_L = 5.3\text{mA}$$

$$I_{EC} = \frac{12 - 0.2}{2.2\text{K}} = 5.3636\text{mA}$$

$$\therefore I_C = I_{EC} - I_S = 0.0636\text{mA}$$



(b) V_{IL} calculation

T_1 is on the verge of turn on $\rightarrow V_B = V_T = 0.5\text{V}$

$$I_2 = \frac{0.5 + 12}{100\text{K}} = 0.125\text{mA}$$

$$I_1 = I_2 = 0.125 = \frac{V_i - 0.5}{15\text{K}}$$

$$\Rightarrow V_i = 2.375\text{V}$$

$$\therefore V_{IL} = 2.375\text{V}, V_{OL} = 0.2\text{V} \rightarrow V_{NL} = V_{iL} - V_{OL} = 2.175\text{V}$$

V_{IH} calculation

$T_1 \rightarrow \text{sat}$, on the verge of saturation and F Active.

$$\therefore \beta_{\text{forced}} \approx \beta_F = 30, V_B = 0.8\text{V}, V_C = 0.2\text{V}$$

$$I_C = \frac{12 - 0.2}{2.2\text{K}} = 5.3636\text{mA}, \beta = \frac{I_C}{I_B} \Rightarrow I_B = \frac{I_C}{30} = 0.1788\text{mA}$$

$$I_2 = \frac{0.8 + 12}{100\text{K}} = 0.128\text{mA}, I_1 = I_2 + I_B = 0.3068\text{mA}$$

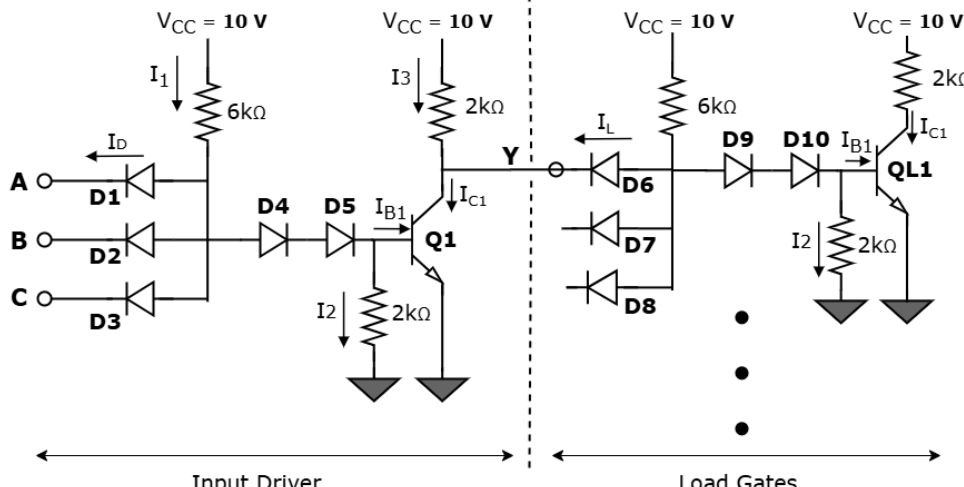
$$I_1 = 0.3068 = \frac{V_i - 0.8}{15\text{K}} \Rightarrow V_i = 5.40\text{V}$$

$$\therefore V_{IH} = 5.40\text{V}, V_{OH} = 10\text{V} \therefore V_{NH} = V_{OH} - V_{IH} = 4.6\text{V}$$

$$\therefore \text{Noise Margin} \rightarrow NM = \min(V_{NL}, V_{NH}) = 2.175\text{V}$$

DTL (ZBH):

Set A

CO2	For the below circuit, $B_F = 30$, $V_{CE}(\text{sat}) = 0.1 \text{ V}$, $V_{BE}(\text{sat}) = 0.8 \text{ V}$  (a) Find I_1, I_2, I_{B1}, I_{C1} and I_D if any input is logical low (0.1 V) for the driver circuit when load gates are not connected. Then also find out the power dissipation (of the driver) for this case. [6 marks] (b) Find I_1, I_2, I_{B1}, I_{C1} and I_D if all input is logical high (10 V) for the driver circuit when load gates are not connected. Then also find out the power dissipation (of the driver) for this case. Prove that Q1 transistor is in Saturation mode. [6+1 marks] (c) The driver circuit is now connected with Load gates. Calculate maximum fanout of the overall DTL setup. [7 marks]	[6+7+7] = 20 marks
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Set A

Ⓐ let's say 'A' is low $\rightarrow V_A = 0.1 \text{ V} \Rightarrow \text{'D}_1 \text{' is conducting}$

so, $V_p = 0.1 + 0.7 = 0.8 \text{ V}$

$\therefore I_1 = \frac{10 - 0.8}{6 \text{ k}} = 1.533 \text{ mA} = I_D$

due to $V_p = 0.8 \text{ V}$;

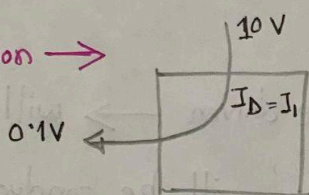
D_4, D_5 turned off

$Q_1 \rightarrow$ cut off

$\therefore I_2 = 0 \text{ mA}$

$I_{B1} = 0 \text{ mA} ; I_{C1} = 0 \text{ mA}$

Power dissipation $\rightarrow$



$P = (10 - 0.1) \times 1.533$
 $= 15.18 \text{ mW}$

Ⓑ $V_A = V_B = V_C = 10 \text{ V} \Rightarrow D_1, D_2, D_3$ all turned off $\Rightarrow \boxed{I_D = 0}$

so, $V_p = 0.7 + 0.7 + 0.8 = 2.2 \text{ V}$

$\therefore I_1 = \frac{10 - 2.2}{6 \text{ k}} = 1.3 \text{ mA}$

$D_4, D_5 \rightarrow$ conducting

$Q_1 \rightarrow$ saturated

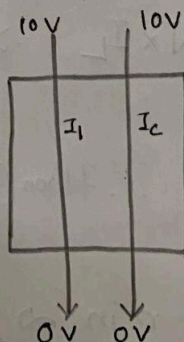
$I_2 = \frac{0.8 - 0}{2 \text{ k}} = 0.4 \text{ mA}$

$\therefore I_{B1} = I_1 - I_2 = (1.3 - 0.4) = 0.9 \text{ mA}$

$I_{C1} = \frac{10 - 0.1}{2 \text{ k}} = 4.95 \text{ mA}$

$\therefore \beta_{\text{forced}} = \frac{I_{C1}}{I_{B1}} = \frac{4.95}{0.9} = 5.5 < \beta_F$ (thus, 'Q1' is in saturation mode)

power dissipation $\rightarrow$



$P = P_1 + P_2$
 $= (10 - 0) \times 1.3 +$
 $(10 - 0) \times 4.95$
 $= 62.5 \text{ mW}$

(c) case $\rightarrow$ any 'input' of the driven low $\rightarrow$ will lead to V_y being 'High' $\rightarrow$ thus 'D₆' will not be conducting, so, $I_L = 0$ mA current flowing into the driven circuit from loads,

so, we can connect as many loads as possible in such case,

$$\text{Fanout} = \infty$$

case $\rightarrow$ all 'inputs' high of the driven $\rightarrow$ will lead to V_y being 'low' $\rightarrow$ thus 'D₆' will be conducting.

For each load, we consider the (011) case, so as a result, Maximum current will flow in 'D₆' path and to our driven circuit,

$$V_y = 0.1, \text{ then, } V_{P_{\text{load}}} = 0.1 + 0.7 = 0.8$$

$$\therefore I_L = \frac{10 - 0.8}{6k} = 1.533 \text{ mA}$$

let's say, there are 'N' numbers of loads connected

$$I_3 = \frac{10 - 0.1}{2k} = 4.95 \text{ mA}$$

$$\text{using constraint rule, } I_{C_{\text{max}}} = \beta_F \times I_{B_1} = 30 \times 0.9 = 27 \text{ mA}$$

$$\therefore I_{C_{\text{max}}} = I_3 + N \times I_L \Rightarrow 27 = 4.95 + N \times 1.533$$

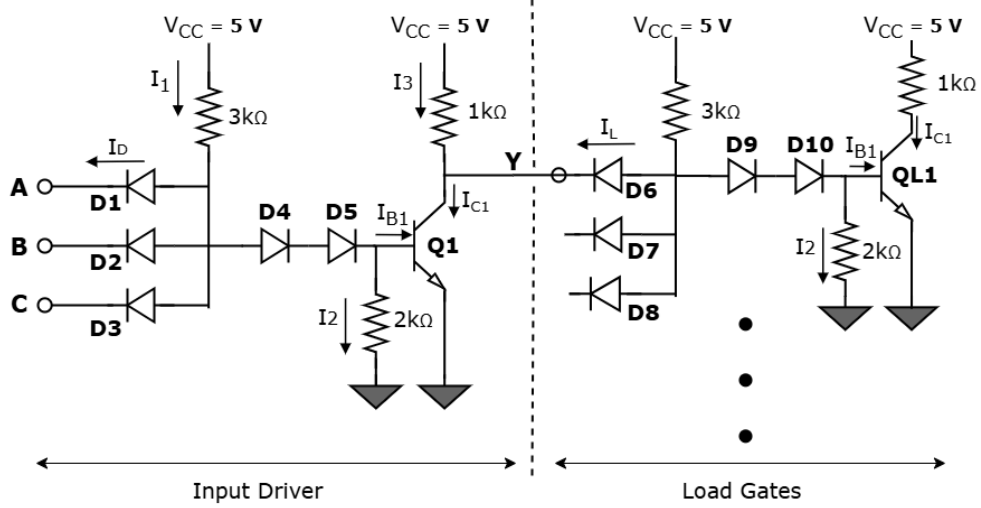
$$\Rightarrow N = 14.38$$

$$\therefore \text{floor}(N) = 14$$

$$\therefore \text{fanout} = 14$$

$$\therefore \text{max}^m \text{ fanout} = \min(\infty, 14) = \boxed{14} \text{ ans}$$

Set B

CO2	For the below circuit, $B_F = 30$, $V_{CE}(\text{sat}) = 0.1 \text{ V}$, $V_{BE}(\text{sat}) = 0.8 \text{ V}$  ← Input Driver Load Gates → (a) Find I_1, I_2, I_{B1}, I_{C1} and I_D if any input is logical low (0.1 V) for the driver circuit when load gates are not connected. Then also find out the power dissipation (of the driver) for this case. [6 marks] (b) Find I_1, I_2, I_{B1}, I_{C1} and I_D if all input is logical high (5 V) for the driver circuit when load gates are not connected. Then also find out the power dissipation (of the driver) for this case. Prove that Q1 transistor is in Saturation mode. [6+1 marks] (c) The driver circuit is now connected with Load gates. Calculate maximum fanout of the overall DTL setup. [7 marks]	[6+7+7] = 20 marks
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selb @ let's say 'A' is low $\rightarrow V_A = 0.1V \Rightarrow$ 'D1' is conducting

so, $V_p = 0.1 + 0.7 = 0.8V$

$\therefore I_1 = \frac{5 - 0.8}{3k} = 1.4 \text{ mA} = I_D$

due to $V_p = 0.8V$;

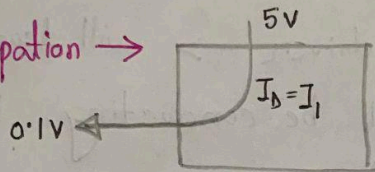
D4, D5 turned off

Q1 $\rightarrow$ cut off

$\therefore I_2 = 0 \text{ mA}$

$I_{B1} = 0 \text{ mA} ; I_{C1} = 0 \text{ mA}$

power dissipation $\rightarrow$



$P = (5 - 0.1) \times 1.4$
 $= 6.86 \text{ mW}$

(b) $V_A = V_B = V_C = 5V \Rightarrow D_1, D_2, D_3$ all are turned off $\Rightarrow I_D = 0$

so, $V_p = 0.7 + 0.7 + 0.8 = 2.2V$

$\therefore I_1 = \frac{5 - 2.2}{3k} = 0.933 \text{ mA}$

D4, D5 $\rightarrow$ conducting

Q1 $\rightarrow$ saturated

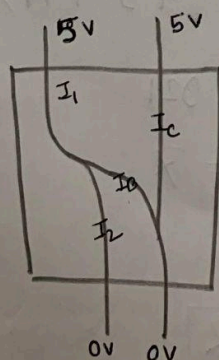
$I_2 = \frac{0.8 - 0}{2k} = 0.4 \text{ mA}$

$\therefore I_{B1} = I_1 - I_2 = (0.933 - 0.4) = 0.533 \text{ mA}$

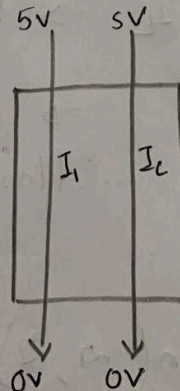
$I_{C1} = \frac{5 - 0.1}{1k} = 4.9 \text{ mA}$

$\therefore \beta_{forced} = \frac{I_{C1}}{I_{B1}} = \frac{4.9}{0.533} = 9.19 < \beta_F$ (thus, 'Q1' is in saturation mode)

power dissipation $\rightarrow$



$\Rightarrow$



$P = P_1 + P_2$

$= (5 - 0) \times 0.933 + (5 - 0) \times 4.9$

$= 29.165 \text{ mW}$

(c) case $\rightarrow$ any 'input' of the driven low $\rightarrow$ will lead to V_y being 'High' $\rightarrow$ thus ' D_6 ' will not be conducting.
 so, $I_L = 0 \text{ mA}$ current flowing into the driven circuit from loads.
 so, we can connect as many loads as possible in such case.

$$\text{Fanout} = \infty$$

case $\rightarrow$ all 'inputs' high of the driver $\rightarrow$ will lead to V_y being 'low' $\rightarrow$ thus ' D_6 ' will be conducting. For each load, we consider the (011) case, so as a result, maximum current will flow in ' D_6 ' path and to our driven circuit.

$$V_y = 0.1, \text{ then } V_{P_{\text{load}}} = 0.1 + 0.7 = 0.8$$

$$\therefore I_L = \frac{5 - 0.8}{3k} = 1.4 \text{ mA}$$

$$I_3 = \frac{5 - 0.1}{1k} = 4.9 \text{ mA}$$

let's say there are ' N ' numbers of loads connected

$$\text{using constraint rule, } I_{c_{\text{max}}} = \beta_F \times I_{B_1} = 30 \times 0.533 = 15.99 \text{ mA}$$

$$\therefore I_{c_{\text{max}}} = I_3 + N \times I_L \Rightarrow 15.99 = 4.9 + N \times 1.4$$

$$\Rightarrow N = 7.921$$

$$\therefore \text{floor}(N) = 7$$

$$\therefore \text{fanout} = 7$$

$$\therefore \text{max}^m \text{ fanout} = \min(\infty, 7) = \boxed{7}_{\text{ans.}}$$

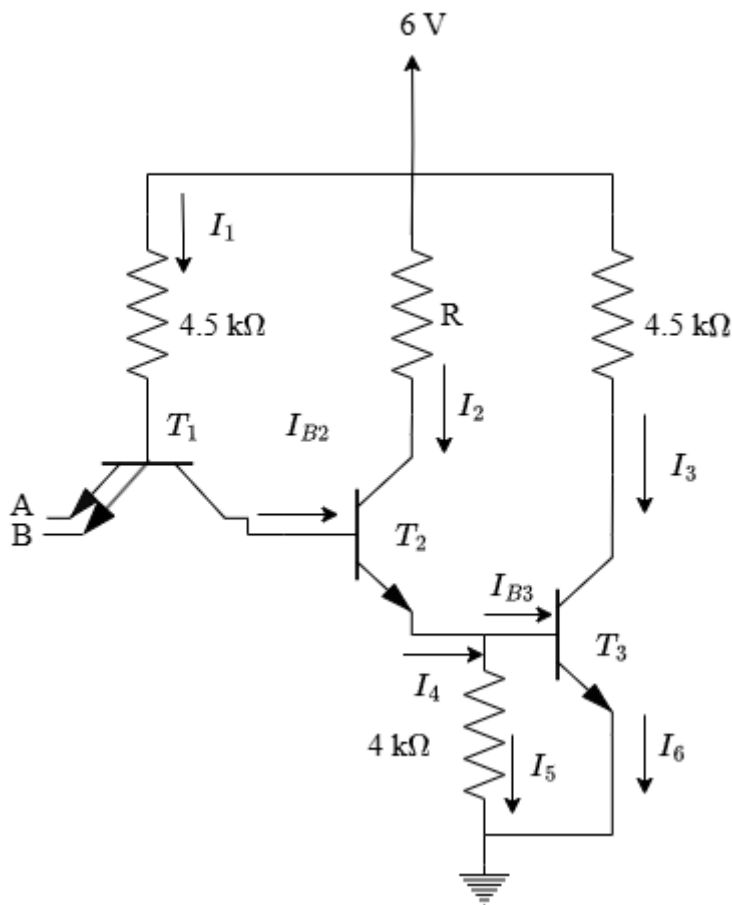
DL & TTL (ZAZ):

Set A

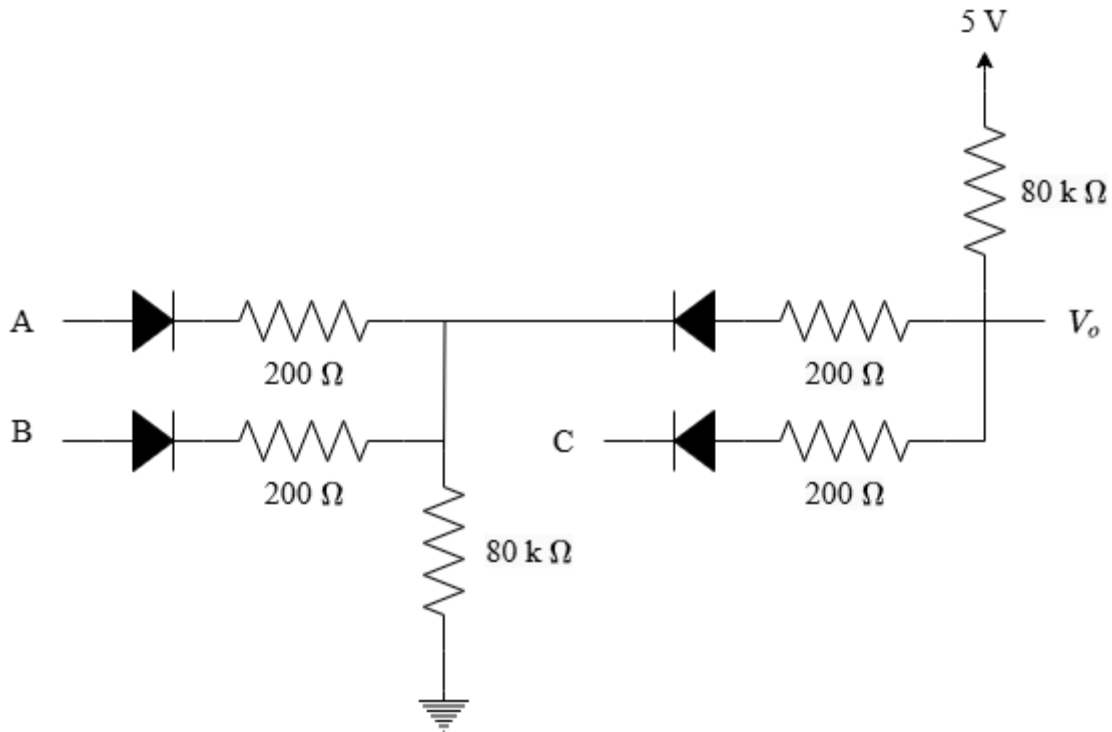
For the given TTL NAND gate,

$\beta_R = 0.2$, $V_{CE}(\text{saturation}) = 0.2\text{ V}$, $V_{BE}(\text{saturation}) = 0.8\text{ V}$, $V_{BC}(\text{reverse active}) = 0.7\text{ V}$

Input voltage, $V_{in}(\text{high}) = 6\text{ V}$, $V_{in}(\text{low}) = 0.2\text{ V}$



(a)	CO2	If $I_6 = 4.74\text{ mA}$ when both inputs are high, find the value of R .	[10]
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For the circuit shown, input voltage, $V_{in}(high) = 5\text{ V}, V_{in}(low) = 0\text{ V}$

(c)	CO2	What logic function does the above circuit implement?	[2]
(b)	CO2	Find the output voltage & power dissipation of the entire circuit when A=low, B=high, C=low	[8]

Solution:

$$a) I_6 = I_3 + I_{B3} = I_3 + I_4 - I_5 = I_3 + I_{B2} + I_2 - I_5$$

$$= I_3 + I_1 + I_{EA} + I_{EB} + I_2 - I_5$$

$$\rightarrow 4.74 = \frac{6-0.2}{4.5} + \frac{(6-2.3)}{4.5} + 0.2 \times \frac{6-2.3}{4.5} \times 2 + \frac{6-1}{R} - \frac{0.8}{4}$$

$$R = 2 \text{ k}\Omega$$

b) $(A + B)C$

c) KCL at output, $\frac{5-V_o}{80} = \frac{V_o-0.7}{0.2} \rightarrow V_o = 0.71 \text{ V}$

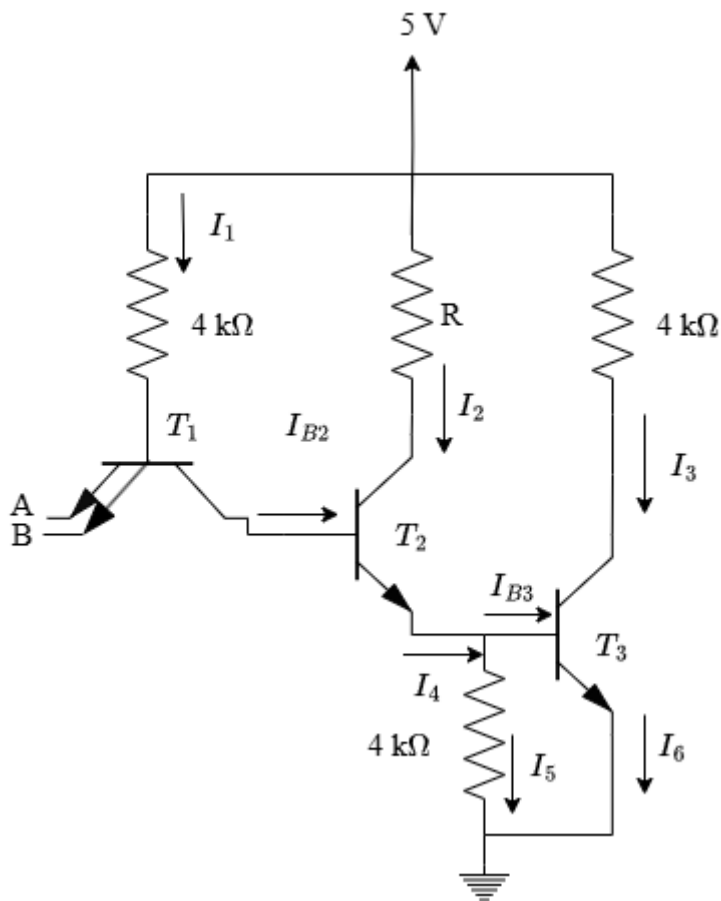
$$P = (5 - 0) \times \frac{5-0.7-0}{0.2+80} + (5 - 0) \times \frac{5-0.71}{80} = 0.536 \text{ mW}$$

Set B

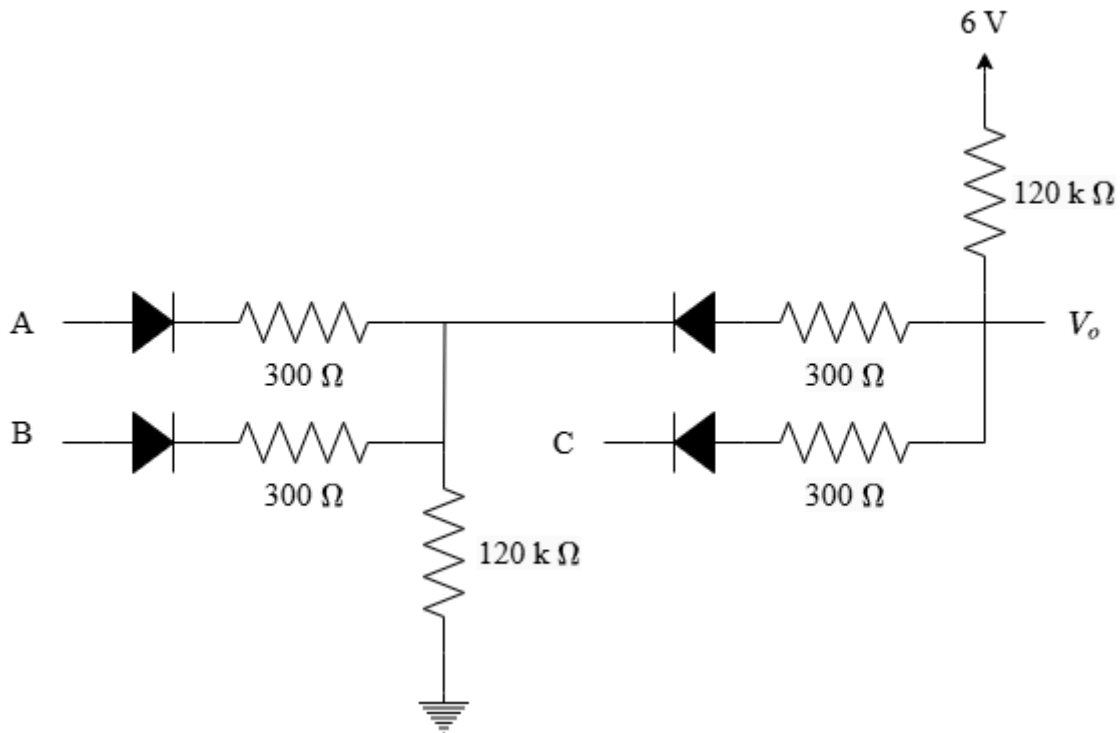
For the given TTL NAND gate,

$$\beta_R = 0.2, V_{CE}(\text{saturation}) = 0.2 \text{ V}, V_{BE}(\text{saturation}) = 0.8 \text{ V}, V_{BC}(\text{reverse active}) = 0.7 \text{ V}$$

Input voltage, $V_{in}(\text{high}) = 5 \text{ V}, V_{in}(\text{low}) = 0.2 \text{ V}$



(a)	CO2	If $I_6 = 4.23 \text{ mA}$ when both inputs are high, find the value of R.	[10]
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For the circuit shown, input voltage, $V_{in}(high) = 6 \text{ V}, V_{in}(low) = 0 \text{ V}$

(c)	CO2	What logic function does the above circuit implement?	[2]
(b)	CO2	Find the output voltage & power dissipation of the entire circuit when A=low, B=high, C=low	[8]

Solution:

$$\begin{aligned}
 \text{a) } I_6 &= I_3 + I_{B3} = I_3 + I_4 - I_5 = I_3 + I_{B2} + I_2 - I_5 \\
 &= I_3 + I_1 + I_{EA} + I_{EB} + I_2 - I_5
 \end{aligned}$$

$$\rightarrow 4.23 = \frac{5-0.2}{4} + \frac{(5-2.3)}{4} + 0.2 \times \frac{5-2.3}{4} \times 2 + \frac{5-1}{R} - \frac{0.8}{4}$$

$$R = 1.75 \text{ k}\Omega$$

b) $(A + B)C$

c) KCL at output, $\frac{6-V_o}{120} = \frac{V_o-0.7}{0.3} \rightarrow V_o = 0.71 \text{ V}$

$$P = (6 - 0) \times \frac{6-0.7-0}{0.3+120} + (6 - 0) \times \frac{6-0.71}{120} = 0.529 \text{ mW}$$